

ALDO ORTEGA YUCRA

aldortega65@gmail.com | alopalao.github.io/Aldo-Ortega | linkedin.com/in/aldortega | github.com/alopalao

EDUCATION

Master of Science in Computer Science

Expected Dec. 2027

Florida International University; Miami, Florida

GPA: 3.87/4.00

WORK EXPERIENCE

Center For Internet Augmented Research & Assessment (CIARA) | Miami, Florida

Jan. 2023 to Present

Research Application Developer Intern

- Led the development of open-source network management tools for Kytos-ng in **Python** and **Linux** so researchers and engineers could manage network traffic reliably.
- Designed reusable web UI components in **Vue.js** so future student developers could use them and save time in styling and interface design decisions.
- Redesigned database structure in **MongoDB** from sparse arrays to **reduce document storage size by over 90%**, so data retrieval and update was made faster and cheaper.
- Upgraded system concurrency with Python asyncio and multi-threading to **handle 10x more processes** (1,000 req/sec), maintaining API stability during peak traffic loads.
- **Mentored 5+ student developers** annually through structured Git and Python code reviews to accelerate onboarding and uphold software quality standards.

Center For Internet Augmented Research & Assessment (CIARA) | Miami, Florida

Sept. 2022 to Dec. 2022

Application Developer Intern

- Constructed and maintained back-end for a repository leveraging **PyMongo** for efficient database management.
- Enforced robust data validation with **Pydantic**, automatically enforcing data format rules and prevents future developers from runtime errors.

Amazon | Austin, Texas

May 2022 to Aug. 2022

Intern Software Developer Engineer

- Built modern, intuitive **React UI** components for customer support tools, simplifying refund and cancellation workflows for agents.
- Replaced large API calls with **GraphQL** queries to fetch order data across internal services without loading unnecessary customer information.

PROJECTS

Moving pieces puzzle | Python | <https://github.com/Alopalao/Moving-pieces-puzzle>

- Built an interactive sliding tile puzzle game in Python using Tkinter, splitting custom images into randomized NxN grids for real-time user gameplay.
- Implemented an automated hint system powered by the A* pathfinding algorithm, calculating best next moves to guide users or auto-solve configurations.

MainRiskApp AI | Python, Vue | <https://github.com/AlexDeleon873/MainRiskApp> | Hackathon winner

- Developed a full-stack emergency response web app in Python (Flask), JavaScript, and AI Vision, facilitating first responders to stream live video feeds and photos for instant AI disaster analysis.

SKILLS

Languages: Python, Vue.js, JavaScript, HTML, CSS, NoSQL, Kotlin, Java, C++

Technologies: Git, Linux, Docker, React, MongoDB, Virtual Machines, Agentic tools.

Technical Competencies: Multi-threading and Asynchronous processes, Object-Oriented Programming, AI leverage.

Extracurricular Involvement: Google Developer Group (GDG at FIU) Club Member